

RepNet Progress Report - April 14, 2026

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1 Introduction

Risk Estimation of Preeclampsia Network (RepNet) is our research in improving the classification of 12-Lead electrocardiogram (ECG) waveforms for estimation of preeclampsia. We aim to do this via a transformer based convolutional neural network (CNN) trained on a proprietary UC Davis Health (UCDH) dataset. We aim to augment dataset imbalance using a transformer based generative adversarial network (GAN). We will also explore the retraining of existing ECG Large Foundation Models (LFM) as well as benchmarking against their performance versus our specific architecture. We aim to bolster clinical interpretability through saliency mapping of the neural network's attention mechanism. And lastly we aim to train a random forrest (RF) classifier and visualize manually extracted ECG features from the RF classifier alongside the attention mapping and saliency of deep learning to augment clinical workflows by providing explainability to clinicians.

2 Review of the Literature

We performed a review of the literature by first querying ChatGPT, which performed broad internet searches over academic archives such as Pubmed, Google Scholar, etc. We queried it for ranking the best papers across specific fields we intend to study, such as the state of the art of machine learning (ML) models for preeclampsia or deep learning for the entire field of ECG classification. Papers were evaluated based on their breadth, rigor, and relevance to a novel research project. After this review of literature reviews, we intend to select the most relevant original studies each mention for our actual deep dive into model architecture and methods whilst doing development and validation, and such will be reported in Progress Report 2. See appendix A for the list of reviewed literature for Progress Report 1.

2.1 Preeclampsia ML / DL Literature Reviews

We sought a comprehensive review over the field machine learning and deep learning research into preeclampsia, to see what work had been done to solve this problem in the past. We selected Özcan et al. as it both a broad overview and review of specific papers and the trade offs of their models. Özcan included 35 papers published from 2018 to 2023 [cite]. It serves as a good primer into the state of the art. We also selected a statistical meta-analysis by Liu et al that goes much deeper into comparison of model performance [cite].

(table showing distribution of model types)

None of the models presented in Özcan et al or Liu et al were trained to classify ECG waveforms using machine learning, they were all various methods of learning from other features such as electronic health record (EHR) data and risk factors [double cite]. To our knowledge there is only one relevant previous work, Butler et al which showcases a 1-D CNN they call ECG-AI [cite]. There is also an older study by Euliano et al which is on 3-Lead instead of

12-Lead ECG and they perform linear discriminant analysis (LDA), a classical ML approach [cite].

By extension, the reviews do not showcase any Random Forest or Decision Trees to classify ECG Waveforms of preeclampsia, though many RF models exist using other features such as EHR data. We further confirm this by an extended-thinking ChatGPT query, asking it to return all ECG Waveform ML papers besides Butler et al and it comes up blank. This indicates that the field of both classical ML and modern DL on ECG waveforms of preeclampsia is an untapped research field that is ripe for contribution.

2.2 ECG Classification ML / DL Lit. Reviews

In the general field of ECG Waveform classification, we selected the literature review by Boulif et al as a broad scope primer that covers both ML and DL methods, and then Prakash et al as a deep systematic review that covers much more research process and model architecture [double cite].

Since we are focusing on Deep Learning such as CNN enhanced by attention mechanisms, we selected we selected Y. Ansari et al. for its deep systematic analysis of DL model architectures, and M. Ansari et al. for its comprehensive survey of transformers and LLMs specifically [double cite].

2.3 ECG Classification Large Foundation Models

Large Foundation Models hold promise for their training on millions of ECG waveforms, which far exceed any purpose-built dataset we can have access to. We selected ECG-FM since it is an "open" model, the authors giving free license as well as providing benchmarking resources [cite]. There is a larger model ECG-Founder but this appears to be a commercial-grade model rather than a research-grade one, and is perhaps beyond the scope of this project and hence we are electing not to use it [cite]

2.3.1 General Overview. Han et al. stands apart from the rest as a review of foundation models for electrocardiograms, moving beyond transformers, llms, into large trained models that are then fine tuned for specific applications

2.3.2 Leading ECG Large Foundation Models. We will experiment with Large Foundation Models, notably ECG-FM and ECG-Founder, and how they compare

- Li <https://github.com/PKUDigitalHealth/ECGFounder>

2.4 Specific Techniques

Lastly we review specific techniques, an introduction to the actual practice of preprocessing and data augmentation that enable models to perform.

2.4.1 ECG Features of Preeclampsia. A particular citation in Özcan et al. pointed to an old paper that was an overview of ECG features

of preeclampsia, and QT interval is noted as highly relevant for preeclampsia

- Abodashy
- Angeli

2.4.2 Preprocessing and Decision Trees on ECG Waveforms. Jia an overview of the data preprocessing techniques to be used for all our models, Satija provides methods for ECG signal quality assessment, and Park et al. decision tree on ecg waveform paper

- Jia
- Satija
- Park et al.

2.5 Data Augmentation

overview of ECG data augmentation by Rahman, and transformer GAN xia et al

2.6 Other Assigned Literature sans Butler et al.

We were provided literature in this capstone assignment that falls into two camps

2.6.1 Machine Learning for Preeclampsia. Xue Attia

2.6.2 Dimensionality Reduction in Leads. Joshi Butler

2.6.3 Decision Tree Classification. Furthermore we were provided a repository of relevant literature, and one that stood out was Musa et al as they performed Principal Component Analysis, a form of dimensionality reduction in linear algebra, on a Decision Tree Classifier for Preeclampsia. which we feel is relevant for our lead reduction and random forest model

3 Butler et al. ECG-AI Architecture and Weights

We have obtained the model dimensions and weights of the ECG-AI model mentioned in Butler et al. from GitHub, it can serve as a benchmark against our own model, both in the performance of their model weights, and the performance of their model architecture trained on our dataset.

Furthermore, the architecture serves as a starting point where we can experiment with transfer learning in freezing in retraining different layers, append, pre-pend, replace, or shim layers to achieve new functions such as attention, all in the aim of even better performance.

However, the datasets utilized by this model remain proprietary, though we have reached out to the associated institutions with a request to access the datasets, one of the emails has bounced and we are waiting for a response from the other.

3.1 Model Architecture

The Butler et al. ECG-AI model is a 1-D residual convolutional neural network that accepts 12-lead ECG signals of length 2250 samples. The architecture comprises six residual blocks organized into three stages of progressively increasing filter width (16, 32, 64), each followed by max-pooling and dropout for regularization. A final flatten and dense softmax layer produces a two-class output. Figure 1 summarizes the architecture.

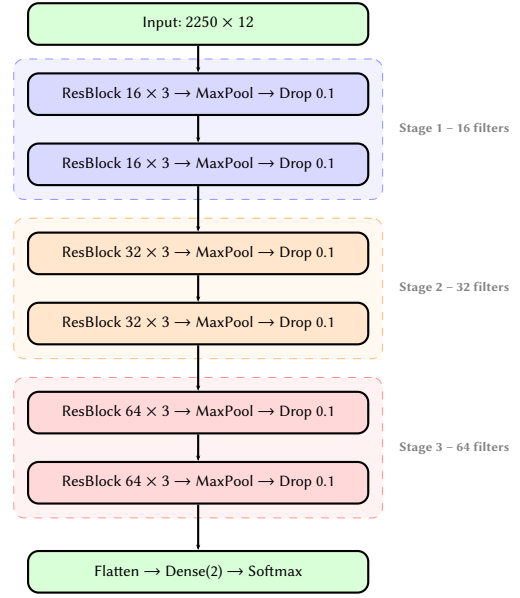


Figure 1: High-level view of the Butler et al. ECG-AI architecture showing six residual blocks across three filter-width stages. See Appendix B for the decomposed layer-by-layer model architecture.

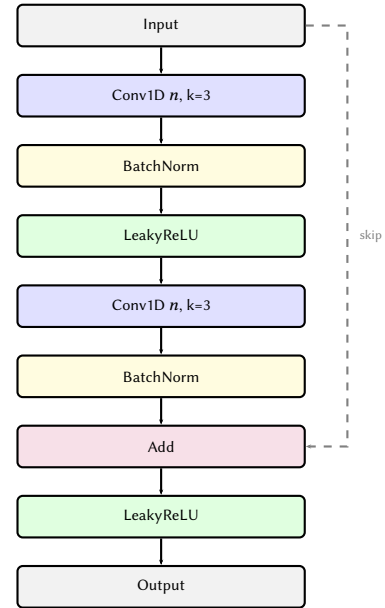


Figure 2: Internal structure of a ResBlock. The skip connection bypasses two Conv1D–BatchNorm–LeakyReLU pairs. A 1×1 projection is applied on the skip path at stage boundaries to match filter dimensions.

A Summary of Reviewed Literature

Table 1: Summary of Reviewed Literature

Paper	Type	Purpose
<i>Preeclampsia - General ML/DL Reviews</i>		
Özcan et al.	Review	General overview of ML models in preeclampsia
Liu et al.	Review	
<i>ECG Classification - General ML/DL Reviews</i>		
Boulif et al.	Review	Broad overview of ECG classification methods
Prakash et al.	Review	Systematic review of ML and DL for ECG beat classification
<i>ECG Classification - DL and Transformers</i>		
Y. Ansari et al.	Review	Overview of DL architectures for ECG (2017-2023)
M. Ansari et al.	Review	Survey of transformers and LLMs for ECG processing
<i>ECG Classification - Large Foundation Models</i>		
Han et al.	Review	Review of large foundation models for ECG
McKeen et al.	Original	ECG-FM open source foundation model
<i>Preeclampsia - ECG Feature Extraction Studies</i>		
Abodashy et al.	Original	ECG features of preeclampsia
Angeli et al.	Original	QT interval relevance for preeclampsia
<i>ECG Classification - Preprocessing Reviews</i>		
Jia et al.	Review	Overview of ECG preprocessing techniques
Satija et al.	Review	ECG signal quality assessment
<i>Preeclampsia - Decision Tree Approach</i>		
Park et al.	Original	Decision tree methods on ECG waveforms
Musa et al.	Original	PCA on decision tree classifier for preeclampsia
<i>ECG Classification - Data Augmentation</i>		
Rahman et al.	Review	Overview of ECG data augmentation
Xia et al.	Original	Transformer GAN for data augmentation
<i>Assigned Literature from Project Assignment</i>		
Butler et al.	Original	1-D CNN for preeclampsia ECG classification
Xue et al.	Original	ML methods for preeclampsia prediction
Attia et al.	Original	ML applications in ECG analysis
Joshi et al.	Original	Lead reduction in ECG analysis

B Butler et al. Full Layer-by-Layer Architecture

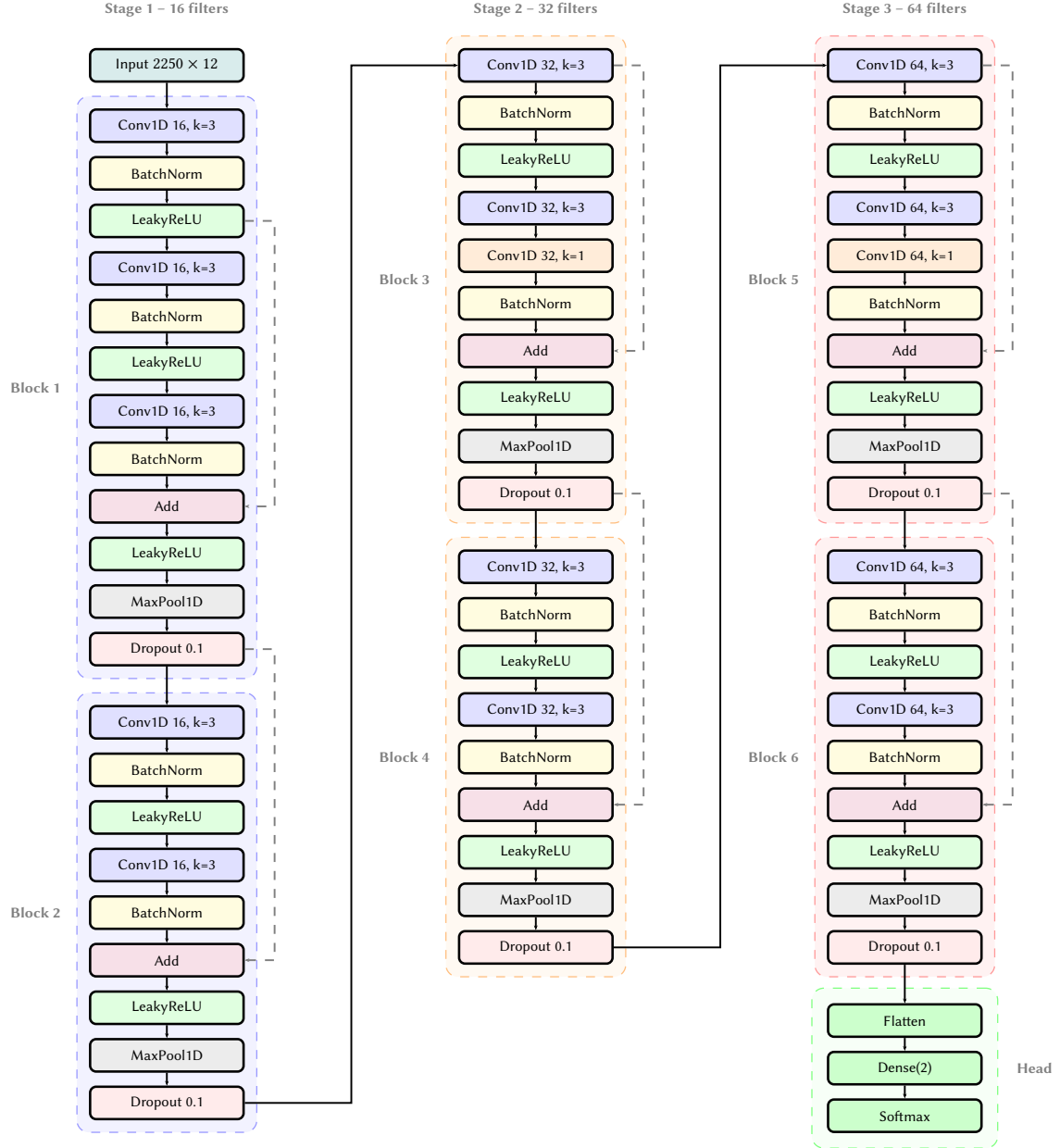


Figure 3: Complete layer-by-layer architecture of the Butler et al. ECG-AI model, expanded from Figure 1. The network comprises six residual blocks across three stages of increasing filter width: Stage 1 (16 filters), Stage 2 (32 filters), and Stage 3 (64 filters). Each layer is shown individually – Conv1D (blue), BatchNorm (yellow), LeakyReLU (green), Add (purple), MaxPool1D (gray), and Dropout (red). Dashed gray arrows indicate residual skip connections that bypass the convolutional layers within each block. Orange blocks denote 1×1 projection convolutions at Blocks 3 and 5, where the skip path requires dimensionality matching due to the increase in filter count. The classification head (Flatten, Dense, Softmax) follows after Stage 3.